

Bertrand Juan

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EDUCATION

The Cooper Union for the Advancement of Science and Art

Expected Graduation: May 2028

Bachelor of Engineering in Mechanical Engineering

Full Tuition Scholarship 2024

Relevant Coursework: Modern Physics, Thermodynamics, Mechanics of Materials, Systems Engineering, Design & Prototype, Principles of Electrical Engineering

Clubs / Activities: Satellite Launch Initiative, ASME, Badminton Club (President), Track Club (Pacer), Musical Club

PROJECTS

Cooper Union Satellite Launch Initiative

New York, NY

Payload Project Lead

Aug. 2025 - Present

- Worked to determine emissivity, absorptivity, and reflectivity of e-paper while vary pixel densities were displayed
- Covered a portion of e-paper with electrical tape and measured the temperatures with an IR camera and thermal couples. Deduced the true emissivity of e-paper by accounting for the difference in measured temperature and known value of electrical tape emissivity.
- Used a solar simulator to heat e-paper and measured its temperature change to determine absorptivity. Steady state determined by measuring a brass cylinder's temperature until it had not changed over a long period of time.
- Created a streamline and continuous system to collect experimental data.
- Abstract accepted to The National Conference of Undergraduate Researchers, wrote documentation, made biweekly progress presentations, and submitted a final report.

Ground Station Crew

Oct. 2024 – May 2025

- Built a stepper motor system to allow antenna to have two degrees of freedom in rotation
- Assembling and disassembling antenna array for storage and testing
- Created compact circuits to control motors that fit on the antenna mount

Flapping Wing Dynamics – ME 200

Aug. 2025 – Dec. 2025

- Created a mathematical model for the oscillatory behavior of a flapping wing mobile
- Used MATLAB to solve Differential-Algebraic Equations accounting for drag and physical constraints
- Made a flapping wing mobile that had smooth rotation about a shaft and symmetrical about it to constrain it to vertical motion. This allowed for more accurate representation of mathematical assumptions in our model.
- Used tracker video analysis to determine the oscillation of constructed mobile to compare to MATLAB model

Wales: Water Based Physical Motion Rectifier – EID 103

Jan. 2025 – May 2025

- Created a system that turned variable speed rotation input into a constant speed rotation output
- Used Onshape to make bevel gears, pulley systems, and Archimedes screws with proper tolerances
- Made block flow diagrams and participate in design sprints in a team of 4 to rapidly come up with new ideas

Task Robot – EID 101

Aug. 2024 – Dec. 2024

- Used IR sensors and a three-wheel drive system to follow lines, solve mazes, and implement a high speed PID control system
- Created a clamp system that could locate objects with an ultrasonic sensor to pick up and place objects
- Wrote code in C++ to control all systems through an Arduino Uno
- Able to minimize the required number of pins to control all systems on the robot

SKILLS:

Programming Languages: Python (NumPy, matplotlib), R, C, C++, MATLAB (ode15i)

Software: Onshape (CAD), AutoCAD, Arduino, Davinci Resolve, Canva, MEGA11

Manufacturing: Woodwork, Hand and Power Tools, Laser Cutting, 3d Printing, Metal Casting

Computer Skills: Microsoft 365, Excel, PowerPoint, Word, Teams, Google Workspace

Languages: English, Chinese (Fluent)